

THIRUNAGARI AUM NAMAHA

Portfolio: portfolio-repo-brown.vercel.app | GitHub: github.com/Aumnamaha

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TECHNICAL PROFILE

Computer Science undergraduate specializing in AI & ML with hands-on experience developing full-stack web applications, AI/ML inference pipelines, and geospatial analysis tools. Proficient in Python, JavaScript, C++, FastAPI, and React, with a strong background in data structures, asynchronous backend workflows, and collaborative open-source engineering. Proven track record in hackathons and structured technical training programs.

TECHNICAL SKILLS

- Programming Languages: Python, C++, C, JavaScript, HTML, CSS
- Frameworks & Libraries: FastAPI, React, Next.js, Node.js, Streamlit, Tailwind CSS, OpenCV, Tkinter
- Data Science & Geospatial: Remote Sensing & GIS (Python), NumPy, Pandas, Data Visualization
- Tools & Platforms: Git & GitHub, VS Code, Jupyter Notebook, Chrome Extension API
- Core Competencies: Asynchronous API Design, Full-Stack Development, Problem Solving, Clean Code & Debugging

EXPERIENCE & TRAINING

AI Engineering Intern | Swecha / VISWAM.AI (IIIT Hyderabad)

June 2026 – July 2026

- Developed open-source AI solutions targeting societal and technological challenges for the Global South.
- Achieved key hackathon project objectives by collaborating on modular, scalable code bases version-controlled on Swecha GitLab infrastructure (code.swecha.org/aumnamaha).

Remote Sensing & GIS Trainee | India Space Academy (ISA)

May 2026 – June 2026

- Completed intensive summer training on spatial data processing, satellite image analysis, and geospatial visualization using Python.
- Implemented Python scripts to handle geospatial datasets and perform automated spatial data manipulation.

EDUCATION

Bachelor of Technology in Computer Science Engineering (AI & ML Specialization)

2023 – 2027

GITAM University, Hyderabad, Telangana

CERTIFICATIONS & ACHIEVEMENTS

- HackWithHyderabad 3.0 Participant: Recognized for problem-solving and rapid application prototyping in an 8-hour hackathon held at Microsoft Office, Hyderabad (October 2026).
- Swecha / VISWAM.AI Internship Certificate: Awarded for completing hackathon project deliverables for impactful AI solutions (July 2026).
- India Space Academy Certificate: Certified in "Remote Sensing & GIS Using Python" under the Department of Space Education, India Space Week (June 2026).

PROJECTS

A11-LLM-OPTIMIZER | PYTHON, PYTORCH, CUDA, DYNAMIC QUANTIZATION, NVME MEMORY-MAPPING

- Architected an experimental runtime offloading engine enabling local execution of 100B+ MoE models on consumer GPUs (4GB VRAM / 16GB RAM).
- Decoupled Dense Attention into GPU VRAM (4.5GB), cached active experts in System RAM (20GB), and mapped a 51B N-Gram lookup table (~28GB) to NVMe SSD using asynchronous mmap zero-copy DMA streaming (<20μs latency per token).
- Optimized Unsloth Dynamic 2.0 Quantization (UD-Q2_K_XL) pipeline for Qwen3.8-Flash-Next, achieving token generation speeds of up to 70+ tokens/sec on consumer hardware.

STUDENT FEEDBACK PORTAL | PYTHON, FLASK, SQLALCHEMY, MYSQL, ROLE-BASED ACCESS CONTROL

- Engineered a modular full-stack academic feedback management portal with multi-role access controls (Student, Faculty, Admin) and session-based authentication.
- Implemented automated escalation and review pipelines featuring deadline countdown timers, Reddit-style nested comment threads, and multi-criteria status filtering.
- Built custom verification workflows with automatic escalation triggers after 3 failed verifications, alongside data aggregation dashboards for department analytics.

SENTRYMIND | COMPUTER VISION, REAL-TIME AI ANALYTICS, EVENT-DRIVEN ARCHITECTURE (HACKATHON PROJECT)

- Designed a high-throughput surveillance and threat-detection platform yielding sub-100ms inference processing times.
- Integrated event-driven alert triggers and dynamic UI overlays to deliver immediate situational awareness for safety monitoring.
- Architected concurrent multi-stream video pipelines to scale processing efficiency across multiple live camera feeds.

FEDTRUSTCHAIN | PYTHON, FEDERATED LEARNING, BLOCKCHAIN, CRYPTOGRAPHY (CAPSTONE PROJECT)

- Designed a decentralized privacy-preserving framework uniting Federated Learning with distributed ledger protocols.
- Enforced end-to-end data security by integrating zero-knowledge verification mechanisms across multi-party node networks.
- Achieved 99.2% model consensus stability while preventing central server data exposure and point-of-failure vulnerabilities.

VAANI | REACT NATIVE, EXPO, WHISPER.RN, LLAMA.RN, SQLITE, C++ BINDINGS

- Developed a privacy-first, fully offline voice-to-structured-data mobile app running on-device Whisper Tiny (~75MB) and Qwen2.5-0.5B-Instruct (~390MB) models.
- Engineered an automated LLM extraction pipeline with a Corrective Prompt Retry Engine and local JSON Schema Validator to extract structured records (expenses, tasks, reminders) into expo-sqlite.
- Implemented a zero-cloud Network Guard blocking external traffic post-onboarding, alongside an interactive benchmarking engine tracking exact inference latency (ms), CPU usage, and RAM consumption.

SMART LIBRARY SYSTEM | PYTHON, FASTAPI, LOCAL LLMS, WHISPER STT, WEASYPRINT

- Built an intelligent voice-powered library system following the Waterfall model, utilizing local speech-to-text (Whisper) and local LLMs (Llama-3/Mistral) for natural language query parsing.
- Developed automated book search integration fetching full-text content from open-source platforms (Open Library, Project Gutenberg, Standard Ebooks).
- Built a PDF compilation and generation pipeline providing complete book downloads across 8 languages (English, Hindi, Telugu, Tamil, etc.).

EXPENSIU | REACT NATIVE, EXPO, LLAMA.RN, QWEN2.5, SQLITE, NATIVE C++

- Built an offline, privacy-first AI personal finance tracker using local LLM inference (Llama.rn + Qwen 2.5-0.5B) and regex algorithms to automatically parse bank SMS/notifications.
- Integrated multi-currency parsing supporting 50+ currencies across global number formats, featuring local AI transaction classification across 10 categories.
- Implemented zero-cloud security architecture with disk-encrypted SQLite databases, biometric launch protection (Face ID/Fingerprint), and OS-level secure keystore integration.

GOV SCHEME CHATBOT | PYTHON, NLP, VECTOR SEARCH, SEMANTIC RETRIEVAL, STREAMLIT

- Engineered an AI conversational assistant helping citizens match eligibility criteria against national government welfare schemes.
- Implemented semantic vector search embeddings to retrieve accurate scheme documentation and application guidelines instantaneously.
- Designed a multi-turn query interface with structured eligibility breakdowns using Streamlit.

TIM THE TIME MANAGER | JAVASCRIPT, CHROME EXTENSION API (MANIFEST V3), SERVICE WORKERS

- Built a lightweight Manifest V3 browser extension tracking active tab usage across major media platforms (YouTube, Instagram, Facebook).
- Engineered background service workers running asynchronous timer loops and DOM content script overlays to trigger periodic health & wellness reminders every 40 minutes.
- Ensured 100% data privacy using local browser storage API with zero external network overhead or background thread latency.